

The New Mirror

VOL. 1, NO. 011

2026-03-30

COVER STORY

On taking good care of your phone

I just finished replacing the micro-USB daughterboard on my Samsung Galaxy S5, which involved taking the phone most of the way apart, doing the replacement, and putting it back together.

Drew DeVault

This inspired me to write about my approach to

maintaining my cell phone. I've had this phone for a while and I have no plans to upgrade - I backed the upcoming Purism phone, but I expect to spend months/years on the software before I'll be using that as my daily driver.

I don't want to be buying a new phone every year. That's a lot of money! Though the technophile in me finds the latest and greatest technology appealing, the thought of doing my own repairs and upkeep on a battle-tested phone is equally interesting. Here are the four things I've found most important in phone upkeep.

id="install-lineageos-or-replicant">Install LineageOS or Replicant

Before I installed CyanogenMod when I bought this phone, I did some prying into the stock ROM to see just how bad it was. It was even worse than I expected! There were literally hundreds of apps and services with scary permissions running in the background that could not be removed. These spy on you, wear down your battery, and slow down your phone over time - another form of planned obsolescence.

My phone is still as fast as the day I got it. It does a great job with everything I ask it to do. The first thing you should do with every new phone is install a third-party ROM - ideally, without Google apps. Stock ROMs suck, get rid of it.

id="insist-on-a-user-replacable-battery">Insist on a user-replacable battery

Non-user-replacable batteries are an obvious form of planned obsolescence. Batteries don't last forever and you should never buy a phone that you cannot replace the battery of. A new battery for my S5 costs 10 bucks. 4 years in, I've replaced mine once and I can hold a charge fine for a couple of days.

id="get-a-case">Get a case

This one is pretty obvious, but I didn't follow this advice at first. I've never broken a screen, so I didn't bother with a case. When I decided I was going to keep this phone for a long time, I went ahead and bought one. It doubles the thickness of my phone but at least I can be sure I'm not going to bust it up when I drop it. It still fits in my pocket comfortably so it's no big deal.

id="attempt-repairs-before-you-buy-a-new-phone">Attempt repairs before you buy a new phone

The past couple of months, my phone's micro-USB3 port started to act up a bit. I would have to wiggle the cable a bit to get it to take, and it could stop charging if I rustled my desk the wrong way. I got a replacement USB daughterboard on Amazon for 6 bucks. Replacing it took an hour, but when removing the screen I broke the connection between my home button and my motherboard - which was only 10 bucks for the replacement, including same day shipping. The whole process was a lot easier than I thought it would be.

Be a smart consumer when you're buying a phone. Insist on the replacable battery and maybe read the iFixit teardown. Take good care of it and it'll last a long time. Don't let consumerism get the better of you!

id="get-a-case">Get a case This one is pretty obvious, but I didn't follow this advice at first.

Drew DeVault

{

FRONT PAGE

ADL Bias and Diversity Training

Scott Johnson (fuzzyblog)

As a contractor for the ADL, I recently went through our official bias and diversity training. I put together my notes on that for a new hire who recently joined us.

Here are some of the discussion questions:

Is it ok for people of a group to use slurs about their own identity?

Should airlines be required to provide multiple seats to obese people ?

Does your work place treat everyone fairly?

Can anyone be successful in the USA?

Does the USA live up to its ideals of racial diversity?

Do women have the same opportunities as men?

Do people of color have the same opportunities as white people?

Peanut butter racism

Blind spot bias you may have bias that you see in others but can't see in yourself

White fragility by d Angelo

MTV micro aggressions

(my personal favorite (this is a Scott thing)- Tea and Consent)

}

{

FEATURES

Drew DeVault

Gemini is a network protocol for exchanging hypertext documents — “hypertext” in the general sense of the word, not with respect to the hypertext markup language (HTML) that web browsers understand. It’s a simple network protocol which allows clients to request hypertext documents (in its own document format, *gemtext*). It is, in some respects, an evolution of Gopher, but more modernized and streamlined.

```
$ openssl s_client -quiet -crlf \ -servername
drewdevault.com \ -connect drewdevault.com:1965 \| awk
{' print "response: " $0 }' gemini://drewdevault.com re-
sponse: 20 text/gemini response: ``ASCII art of a rocket
next to "Drew DeVault" in a stylized font response: ^
response: || _____ _ . _ _ response:
||\_____\_____--\_\\\|_____|
|_/_ response: /|||\_ _ \_ \\V\\| |\_ _ Y^__\| |
\|\_ response: :|:|\ ` \| \_ ^ / | ` \_ ^ // _ \| | | |
response: |:|:/_____/___> \_//_____^__>\_/
(____/____/____/___ response: /||\ V V V V V response: **
response: ** response: `` [...]
```

My disdain for web browsers is well documented ¹. Web browsers are extraordinarily complex, and any attempt to build a new one would be a Sisyphean task. Successfully completing that implementation, if even possible, would necessarily produce a Lovecraftian mess: unmaintainable, full of security vulnerabilities, with gigabytes in RAM use and hours in compile times. And given that all of the contemporary web browsers that implement a sufficiently useful subset of web standards are ass and getting assier, what should we do?

something different, like Gemini. Gemini does not solve all of the web's problems, but it addresses a subset of its use-cases better than the web does, and that excites me. I want to discard the parts of the web that Gemini does better, and explore other solutions for anything that's left of the web which is worth keeping (hint: much of it is not).

It's dead simple. A client or server implementation can be written from scratch by a single person in the space of an afternoon or two. A new web browser could take hundreds of engineers millions of hours to complete.

It's opinionated about document formatting. There are no inline links (every link goes on its own line), no formatting, and no inline images. Gemini strictly separates the responsibility of content and presentation. Providing the content is the exclusive role of the server, and providing the presentation is the exclusive role of the client. There are no stylesheets and authors have very little say in how their content is presented. It's still possible for authors to express themselves within these constraints — as with any other constraints — but it allows clients to be simpler and act more as user agents than vendor agents.

Some people argue that what we should have is “the web, but less of it”, i.e. a “sane” subset of web standards. I don’t agree (for one, I don’t think there is a “sane” subset of those standards), but I’ll save that for another blog post. Gemini is a new medium, and it’s different from the web. Anyone who checking it out should be prepared for that and open to working within its constraints. Limitations breed creativity!

For my part, I have been working on a number of Gemini projects. For one, this blog is now available on Gemini , and I have started writing some Gemini-exclusive content for it. I've also written some software you're welcome to use:

libgmni , gmni , and gmnln are my suite of Gemini client software, all written in C11 and only depending on a POSIX-like system and OpenSSL. libgmni is a general-purpose Gemini client library with a simple interface . gmni is a cURL-like command line tool for performing Gemini requests. Finally, gmnln is a line-mode browser with a rich feature-set. Together these tools weigh just under 4,000 lines of code, of which about 1,600 are the URL parser from cURL vendored in.

gmnsrv is a high-performance Gemini server, also written in C11 for POSIX systems with OpenSSL. It supports zero-configuration TLS, CGI scripting, auto-indexing, regex

}

{

Rust for Linux revisited

Drew DeVault

Ugh. Drew's blogging about Rust again.

– You

I promise to be nice.

Two years ago, seeing the Rust-for-Linux project starting to get the ball rolling, I wrote “Does Rust belong in the Linux kernel?”, penning a conclusion consistent with Betteridge's law of headlines. Two years on we have a lot of experience to draw on to see how Rust-for-Linux is actually playing out, and I'd like to renew my thoughts with some hindsight – and more compassion. If you're one of the Rust-for-Linux participants burned out or burning out on this project, I want to help. Burnout sucks – I've been there.

The people working on Rust-for-Linux are incredibly smart, talented, and passionate developers who have their eyes set on a goal and are tirelessly working towards it – and, as time has shown, with a great deal of patience. Though I've developed a mostly-well-earned reputation for being a fierce critic of Rust, I do believe it has its place and I have a lot of respect for the work these folks are doing. These developers are ambitious and motivated to make an impact, and Linux is undoubtedly the highest-impact software in the world, and in theory Linux is enthusiastically ready to accept motivated innovators into its fold to facilitate that impact.

At least in theory. In practice, the Linux community is the wild wild west, and sweeping changes are infamously difficult to achieve consensus on, and this is by far the broadest sweeping change ever proposed for the project. Every subsystem is a private fiefdom, subject to the whims of each one of Linux's 1,700+ maintainers, almost all of whom have a dog in this race. It's herding cats: introducing Rust effectively is one part coding work and ninety-nine parts political work – and it's a lot of coding work. Every subsystem has its own unique culture and its own strongly held beliefs and values.

The consequences of these factors is that Rust-for-Linux has become a burnout

machine. My heart goes out to the developers who have been burned in this

project. It's not fair. Free software is about putting in the work, it's a

classical do-ocracy... until it isn't, and people get hurt. In spite of my

critiques of the project, I recognize the talent and humanity of everyone

involved, and wouldn't have wished these outcomes on them. I also have sympathy

for many of the established Linux developers who didn't exactly want this on

their plate... but that's neither here nor there for the purpose of this post,

and any of those developers and their fiefdoms who went out of their way to make

life difficult for the Rust developers above and beyond what was needed to

ensure technical excellence are accountable for these shitty outcomes. 1

So where do we go now?

Well, let me begin by re-iterating something from my last article on the subject: “I wish [Rust-for-Linux] the best of luck and hope to see them succeed”. Their path is theirs to choose, and though I might advise a moment to rest before diving headfirst into this political maelstrom once again, I support you in your endeavours if this is what you choose to do. Not my business. That said, allow me to humbly propose a different path for your consideration.

Here's the pitch: a motivated group of talented Rust OS developers could build a Linux-compatible kernel, from scratch, very quickly, with no need to engage in LKML politics. You would be astonished by how quickly you can make meaningful gains in this kind of environment; I think if the amount of effort being put into Rust-for-Linux were applied to a new Linux-compatible OS we could have something production ready for some use-cases within a few years.

Novel OS design is hard: projects like Redox are working on this, but it will take a long time to bear fruit and research operating systems often have to go back to the drawing board and make major revisions over and over again before something useful and robust emerges. This is important work – and near to my heart – but it's not for everyone. However, making an OS which is based on a proven design like Linux is much easier and can be done very quickly. I worked on my own novel OS design for a couple of years and it's still stuck in design hell and badly in need of being rethought; on the other hand I wrote a passable Unix clone alone in less than 30 days.

Rust is a great fit for a large monolithic kernel design like Linux. Imagine

having the opportunity to implement something like the dcache from scratch in

Rust is a great fit for a large monolithic kernel design like Linux.

Drew DeVault

Rust, without engaging with the politics – that's something a small group of

}

{

You don't need jQuery

Drew DeVault

It's true. You really don't need jQuery. Modern web browsers can do most of what you want from jQuery, without jQuery.

For example, take MediaCrush . It's a website I spent some time working on with a friend. It's actually quite sophisticated - drag-and-drop uploading, uploading via a hidden form, events wired up to links and dynamically generated content, and ajax requests/file uploads, the whole she-bang. It does all of that without jQuery. It's open source , if you're looking for a good example of how all of this can be used in the wild.

Let's walk through some of the things you like jQuery for, and I'll show you how to do it without.

id="document-querying-with-css-selectors">Document Querying with CSS Selectors

You like jQuery for selecting content. I don't blame you - it's really cool. Here's some code using jQuery:

Now, here's how you can do it on vanilla JS:

You want to make requests in JavaScript. This is how you POST with jQuery:

Here's the same code, without jQuery:

```
class="highlight"> var xhr = new XMLHttpRequest (); // A little deceptively named xhr . open ( 'POST' , '/path/to/endpoint' ); xhr . onload = function () { alert ( this . responseText ); }; var formData = new FormData (); formData . append ( 'parameter' , value ); formData . append ( 'other-Parameter' , value ); xhr . send ( formData );
```

Documentation: XMLHttpRequest

Also a bit more verbose than jQuery, but much simpler than you might've expected. Now here's the real kicker: It works in IE 7, and IE 5 with a little effort. IE actually pioneered XHR.

id="animations">Animations

This is where it starts to get more subjective and breaks backwards compatibility. Here's my opinion on the matter of transitions: dropping legacy browser support for fancy animations is acceptable. I don't think it's a problem if your website isn't pretty and animated on older browsers. Keep that in mind as we move on.

I want to animate the opacity of a .foobar when you hover over it. With jQuery:

```
class="highlight"> $( '.foobar' ). on ( 'mouseenter' , function () { $( this ). animate ( { opacity : 0.5 } , 2000 ); }). on ( 'mouseleave' , function () { $( this ). animate ( { opacity : 1 } , 2000 ); });
```

Without jQuery, I wouldn't do this in Javascript. I'd use the magic of CSS animations:

Documentation: CSS animations

Much better, eh? Works in IE 10+. You can do much more complicated animations with CSS, but I can't think of a good demo, so that's an exercise left to the reader.

id="tree-traversal">Tree traversal

jQuery lets you navigate a tree pretty easily. Let's say you want to find the container of a button and remove all .foobar elements underneath it, upon clicking the button.

Nice and succinct. I'm sure you can tell the theme so far - the main advantage of jQuery is a less verbose syntax. Here's how it's done without jQuery:

```
class="highlight"> document . getElementById ( 'mybutton' ). addEventListener ( 'click' , function () { var foobars = this . parentElement . querySelectorAll ( '.foobar' ); for ( var i = 0 ; i
```

In conclusion

jQuery is, of course, based on JavaScript, and as a result, anything jQuery can do can be done without jQuery. Feel free to ask me if you're curious about how I'd do something else without jQuery.

I feel like adding jQuery is one of the first things a web developer does to their shiny new website. It just isn't really necessary in this day and age. That extra request, 91kb, and load time are probably negligible, but it's still a little less clean than it could be. There's no need to go back and rid all of your projects of jQuery, but I'd suggest that for your next one, you try to do without. Keep MDN open in the next tab over and I'm sure you'll get through it fine.

{

Please don't use Discord for FOSS projects

Drew DeVault

Six years ago, I wrote a post speaking out against the use of Slack for the instant messaging needs of FOSS projects. In retrospect, this article is not very good, and in the years since, another proprietary chat fad has stepped up to bat: Discord. It's time to revisit this discussion.

In short, using Discord for your free software/open source (FOSS) software project is a very bad idea. Free software matters — that's why you're writing it, after all. Using Discord partitions your community on either side of a walled garden, with one side that's willing to use the proprietary Discord client, and one side that isn't. It sets up users who are passionate about free software — i.e. your most passionate contributors or potential contributors — as second-class citizens.

By choosing Discord, you also lock out users with accessibility needs, for whom

the proprietary Discord client is often a nightmare to use. 1 Users who cannot

afford new enough hardware to make the resource-intensive client pleasant to use

are also left by the wayside. Choosing Discord is a choice that excludes poor

and disabled users from your community. Users of novel or unusual operating

systems or devices (i.e. innovators and early adopters) are also locked out of

the client until Discord sees fit to port it to their platform. Discord also

declines service to users in countries under US sanctions, such as Iran.

Privacy-conscious users will think twice before using Discord to participate in

your project, or will be denied outright if they rely on Tor or VPNs. All of

these groups are excluded from your community.

These problems are driven by a conflict of interest between you and Discord. Ownership over your chat logs, the right to set up useful bots, or to moderate your project's space according to your discretion; all of these are rights reserved by Discord and denied to you. The FOSS community, including users with accessibility needs or low-end computing devices, are unable to work together to innovate on the proprietary client, or to build improved clients which better suit their needs, because Discord insists on total control over the experience. Discord seeks to domesticate its users,

where FOSS treats users as peers and collaborators. These ideologies are fundamentally in conflict with one another.

You are making an investment when you choose to use one service over another.

When you choose Discord, you are legitimizing their platform and divesting from

FOSS platforms. Even if you think they have a bigger reach and a bigger

audience, 2 choosing them is a short-term, individualist play which signals a

lack of faith in and support for the long-term goals of the FOSS ecosystem as a

whole. The FOSS ecosystem needs your investment. FOSS platforms generally don't

have access to venture capital or large marketing budgets, and are less willing

to use dark patterns and predatory tactics to secure their market segment. They

need your support to succeed, and you need theirs. Why should someone choose to

use your FOSS project when you refused to choose theirs? Solidarity and mutual

support is the key to success.

There are great FOSS alternatives to Discord or Slack. SourceHut has been investing in IRC by building more accessible services like chat.sr.ht . Other great options include Matrix and Zulip . Please consider these services before you reach for their proprietary competitors.

Perceptive readers might have noticed that most of these arguments can be generalized. This article is much the same if we replace "Discord" with "GitHub", for instance, or "Twitter" or "YouTube". If your project depends on proprietary infrastructure, I want you to have a serious discussion with your collaborators about why. What do your choices mean for the long-term success of your project and the ecosystem in which it resides? Are you making smart investments, or just using tools which are popular or that you're already used to?

If you use GitHub, consider SourceHut 3 or Codeberg . If you use Twitter, consider Mastodon instead. If you use YouTube, try PeerTube . If you use Facebook... don't.

Your choices matter. Choose wisely.

id="fn:1">

Discord had to be sued to take this seriously. Updated at 2021-12-28 15:00 UTC: I asked a correspondent of mine who

}

{

Sourcehut's spartan approach to web design

Drew DeVault

Sourcehut is known for its brutalist design, with its mostly shades-of-gray appearance, conservative color usage, and minimal distractions throughout. This article aims to share some insights into the philosophy that guides this design, both for the curious reader and for the new contributor to the open-source project.

The most important principle is that `sr.ht` is an engineering tool first and foremost, and when you're there it's probably because you're in engineering mode. Therefore, it's important to bring the information you're there for to the forefront, and minimize distractions. In practice, this means that the first thing on any page to grab your attention should be the thing that brought you there. Consider the source file view on `git.sr.ht`. For reference, here are similar pages on GitHub and Gitlab.

The vast majority of the `git.sr.ht` page is dedicated to the source code we're

reading here, and it's also where most of the colors are. Your eye is drawn

straight to the content. Any additional information we show on this page is

directly relevant to the content: breadcrumbs to other parts of the tree, file

mode & size, links to other views on this repository. The nav can take you away

from this page, but it's colored a light grey to avoid being distracting and

each link is another engineering tool - no marketing material or fluff. Contrast

with GitHub: a large, dark, attention grabbing navbar with links to direct you

away from the content and towards marketing pages. If you're logged out, you get

a giant sign-up box which pushes the content halfway off the page. Colors here

are also distracting: note the large line of colorful avatars that catches your

eye despite almost certainly being unrelated to your purpose on this page.

Colors are used much more conservatively on sourcehut. If you log into

`builds.sr.ht` and visit the index page, you're greeted with a large blue "submit

manifest" button, and very little color besides. This is probably why you were

here - so it's made obvious and colorful so your eyes can quickly find it and

get on with your work. Other pages are similar: the `todo.sr.ht` tracker page has

a large form with a blue "submit" button for creating a new ticket, email views

on `lists.sr.ht` have a large blue "reply to thread" button, and

`man.sr.ht` has a large green button enticing new users

towards the tutorials. Red is also used throughout for dangerous actions, like

deleting things. Each button also is unambiguous and relies on the text within

itself rather than the text nearby: the `git.sr.ht` repository deletion page uses

"Delete \$reponame", rather than "Continue".

The last important point in sourcehut's design is the use of icons, or rather the lack thereof. Icons are used extremely conservatively on `sr.ht`. Interactive icons (things you are expected to click) are never shown without text that clarifies what happens when you click them. Informational icons usually have a tooltip which explains their meaning, and are quite rare - only used in cases where real estate limits the use of text. Assigning an icon to every action or detail is not necessary and would add more distractions to the screen.

I'm not a particularly skilled UI designer, so keeping it simple like this also helps to make a reasonably nice UI attainable for an engineer-oriented developer like me. Adding new pages is generally easy and requires little thought by applying these basic principles throughout, and the simple design doesn't take long to execute on. It's not perfect, but I like it and I've received positive feedback from my users.

Red is also used throughout for dangerous actions, like deleting things.

Drew DeVault

}

{

PlugMem: Transforming raw agent interactions into reusable knowledge

Ke Yang, Michel Galley, Chenglong Wang, Jianfeng Gao, Jiawei Han, ChengXiang Zhai · Microsoft Research Blog

Today’s AI agents store long interaction histories but struggle to reuse them effectively.

Raw memory retrieval can overwhelm agents with lengthy, low-value context.

PlugMem transforms interaction history into structured, reusable knowledge.

A single, general-purpose memory module improves performance across diverse agent benchmarks while using fewer memory tokens.

It seems counterintuitive: giving AI agents more memory can make them less effective. As interaction logs accumulate, they grow large, fill with irrelevant content, and become increasingly difficult to use.

More memory means that agents must search through larger volumes of past interactions to find information relevant to the current task. Without structure, these records mix useful experiences with irrelevant details, making retrieval slower and less reliable. The challenge is not storing more experiences, but organizing them so that agents can quickly identify what matters in the moment.

In our recent paper “PlugMem: A Task-Agnostic Plugin Memory Module for LLM Agents,” we introduce a plug-and-play memory system that transforms raw agent interactions into reusable knowledge. Rather than treating memory as text to retrieve, PlugMem organizes that history into structured knowledge designed to support decisions as the agent acts.

Cognitive science offers a useful framework here. It distinguishes between remembering events, knowing facts, and knowing how to perform tasks. Past events provide context, but effective decisions rely on the facts and skills extracted from those events.

This distinction motivated a shift in how we decided to design memory for AI agents. PlugMem implements this shift by converting the agent’s interaction history, such as dialogues, documents, and web sessions, into structured, compact knowledge units that can be reused across tasks.

A key difference between PlugMem and conventional AI memory systems is what gets stored. Traditional approaches store text chunks or named entities (references to people, places, and concepts). PlugMem uses facts and reusable skills as the fundamental building blocks of memory. This design reduces redundancy, increases information density, and improves retrieval precision. It’s built around three core components:

Structure. Raw interactions are standardized and transformed into propositional knowledge (facts) and prescriptive knowledge (reusable skills). These knowledge units are

organized into a structured memory graph, enabling knowledge to be stored in a form designed for reuse.

Retrieval. Rather than retrieving long passages of text, PlugMem retrieves knowledge units that are aligned with the current task. High-level concepts and inferred intents serve as routing signals, surfacing the most relevant information for the decision at hand.

Reasoning. Retrieved knowledge is distilled into concise, task-ready guidance before being passed to the base agent, ensuring that only decision-relevant knowledge enters the agent’s context window.

Figure 1 illustrates how these components work together.

Figure 1. PlugMem organizes different types of agent interactions into a knowledge-centric memory graph, enabling structured retrieval and reasoning.

Most AI memory systems are built for one job. A conversational memory module is designed around dialogue. A knowledge-retrieval system is tuned to look up facts. A web agent’s memory is optimized for navigating pages. Each performs well in its target setting but rarely transfers without significant redesign.

PlugMem takes a different approach. It is a foundational memory layer that can be attached to any AI agent without needing to modify it for a specific task.

To test PlugMem, we evaluated the same memory module on three benchmarks that each make different demands on memory:

Answering questions across long multi-turn conversations

Finding facts that span multiple Wikipedia articles

Making decisions while browsing the web

Across all three, PlugMem consistently outperformed both generic retrieval methods and task-specific memory designs while allowing the AI agent to use significantly less memory token budget in the process.

We wanted to evaluate whether the right information was reaching the agent at the right moment, without overwhelming the model’s context window, which has limited capacity. To do this, we introduced a metric that measures how much useful, decision-relevant information a memory module contributes relative to how much context it consumes.

When we plotted utility against context consumption, PlugMem consistently came out ahead: it delivered more decision-relevant information while consuming less of the AI agent’s context than other approaches, as shown in Figure 2. These results suggest that transforming experience into knowledge—rather than storing and retrieving raw logs—produces memory that is more useful and efficient.

}

{

It's not okay to pretend your software is open source

Drew DeVault

Unfortunately, I find myself writing about the Commons Clause again. For those not in the know, the Commons Clause is an addendum designed to be added to free software licenses. The restrictions it imposes (you cannot sell the software) makes the resulting franken-license nonfree. I'm not going to link to the project which brought this subject back into the discussion - they don't deserve the referral - but the continued proliferation of software using the Commons Clause gives me reason to speak out against it some more.

One of my largest complaints with the Commons Clause is that it hijacks language used by open source projects to proliferate nonfree software, and encourages software using it to do the same. Instead of being a new software license, it tries to stick itself onto other respected licences - often the Apache 2.0 license. The name, "Commons Clause", is also disingenuous, hijacking language used by respected entities like Creative Commons. In truth, the Commons Clause serves to remove software from the commons¹. Combining these problems gives you language like "Apache+Commons Clause", which is easily confused with [Apache Commons][apache-commons].

Projects using the Commons Clause have also been known to describe their license as "permissive" or "open", some even calling their software "open source". This is dishonest. FOSS refers to "free and open source software". The former, free software, is defined by the free software definition, published by GNU. The latter, open source software, is defined by the open source definition, published by the OSI. Their definitions are very similar, and nearly all FOSS licenses qualify under both definitions. These are unambiguous, basic criteria which protects developers, contributors, and users of free and open source software. These definitions are so basic, important and well-respected that dismissing them is akin to dismissing climate change.

Claiming your software is open source, permissively licensed, free software, etc, when you use the Commons Clause, is lying. These lies are pervasive among users of the Commons Clause. The page listing Redis Modules, for example, states that only software under an OSI-approved license is listed. Six of the modules there are using nonfree licenses, and antirez seems content to ignore the problem until we forget about it. They're in for a long wait - we're not going to forget about shady, dishonest, and unethical companies like Redis Labs.

I don't use nonfree software², but I'm not going to sit here and tell you not to make nonfree software. You have every right to license your work in any way you choose. However, if you choose not to use a FOSS license, you need to own up to it. Don't pretend that your software is something it's not. There are many benefits to being a member of the free software community, but you are not entitled to them if your software isn't. This behavior has to stop.

Finally, I have some praise to offer. Dgraph was briefly licensed under Apache plus the Commons Clause, and had the sort of misleading and false information this article decries on their marketing website, docs, and so on. However, they've rolled it back, and Dgraph is now using the Apache 2.0 license with no modifications. Thank you!

id="fn:1">

This is why I often refer to it as the "Anti-Commons Clause", though I felt that was a bit too Stallman-esque for this article. [apache-commons]: <http://commons.apache.org/> ↩

id="fn:2">

Free as in freedom, not as in free beer. ↩

}

{

Spooky action at a distance

Drew DeVault

Einstein famously characterized the strangeness of quantum mechanics as “spooky action at a distance”, which, if I had to pick one phrase about physics to be my favorite, would be a strong contender. I like to relate this to programming language design: there are some language features which are similarly spooky. Perhaps the most infamous of these is operator overloading. Consider the following:

```
x + y
```

If this were written in C, without knowing anything other than the fact that this code compiles correctly, I can tell you that `x` and `y` are numeric types, and the result is their sum. I can even make an educated guess about the CPU instructions which will be generated to perform this task. However, if this were a language with operator overloading... who knows? What if `x` and `y` are some kind of some Vector class? It could compile to this:

```
Vector::operator_plus(x, y)
```

The performance characteristics, consequences for debugging, and places to look for bugs are considerably different than the code would suggest on the surface. This function call is the “spooky action” — and the distance between the “+” operator and the definition of its behavior is the “distance”.

Also consider if `x` and `y` are strings: maybe “+” means concatenation? Concatenation often means allocation, which is a pretty important side-effect to consider. Are you going to thrash the garbage collector by doing this? Is there a garbage collector, or is this going to leak? Again, using C as an example, this case would be explicit:

```
char *new = malloc(strlen(x) + strlen(y) + 1); strcpy(new, x);
strcat(new, y);
```

If the filename of the last file you had open in your text editor ended in `.rs`, you might be frothing at the mouth after reading this code. Strictly for the purpose of illustrating my point, however, consider that everything which happens here is explicit, opt-in to the writer, and obvious to the reader.

That said, C doesn’t get off scott-free in this article. Consider the following code:

```
int x = 10, y = 20; int z = add(x, y); printf(“%d + %d = %d\n”,
x, y, z);
```

You may expect this to print out `10 + 20 = 30`, and you would be forgiven for your naivety.

```
$ cc -o test test.c $ ./test 30 + 20 = 30
```

The savvy reader may have already figured out the catch: `add` is not a function.

```
#define add(x, y) x += y
```

The spooky action is the mutation of `x`, and the distance is between the apparent “callsite” and the macro definition. This is spooky because it betrays the reader’s expectations: it looks and smells like a function call, but it does something which breaks the contract of function calls. Some languages do this better, by giving macros an explicit syntax like `name!(args...)`, but, personally, I still don’t like it.

Language features like this are, like all others, a trade-off. But I’m of the opinion that this trade is unwise: you’re wagering readability, predictability, debuggability, and more. These features are toxic to anyone seeking stable, robust code. They certainly have no place in systems programming.

}

{

China

Drew DeVault

This article will be difficult to read and was difficult to write. I hope that you can stomach the uncomfortable nature of this topic and read my thoughts in earnest. I usually focus on technology-related content, but at the end of the day, this is my personal blog and I feel that it would betray my personal principles to remain silent. I've made an effort to provide citations for all of my assertions.

Note: if you are interested in conducting an independent review of the factuality of the claims expressed in this article, please contact me .

The keyboard I'm typing these words into bears "Made in China" on the bottom. The same is true of the monitor I'm using to edit the article. It's not true of all of my electronics — the graphics processing unit which is driving the monitor was made in Taiwan 1 and my phone was made in Vietnam. 2 Regardless, there's no doubt that my life would be, to some degree, worse off if not for trade with China. Despite this, I am prepared to accept the consequences of severing economic relations with China.

How bad would being cut-off from China's economy be? We're a net importer from China, and by over 4 times the volume. 3 Let's assume, in the worst case, trade ties were completely severed. The United States would be unable to buy \$155B worth of electronics, which we already have domestic manufacturing capabilities for 4 and which have a productive life of several years. We could definitely stand to get used to repairing and reusing these instead of throwing them out. We'd lose \$34B in mattresses and furniture — same story. The bulk of our imports from China are luxury goods that we can already make here at home 5 — it's just cheaper to buy them from China. But cheaper for whom?

This gets at the heart of the reason why we're tied to China economically. It's economically productive for the 1% to maintain a trade relationship with China. The financial incentives don't help any Americans, and in fact, most of us are hurt by this relationship. 6 Trade is what keeps us shackled to the Chinese Communist Party government, but it's not beneficial to anyone but those who are already obscenely rich, and certainly not for our poorest — who, going into 2020, are as likely to be high school dropouts as they are to be doctors. 7

So, we can cut off China. Why should we? Let's lay out the facts: China is conducting human rights violations on the largest scale the world has seen since Nazi Germany. China executes political prisoners 8 and harvests their organs for transplant to sick elites on an industrial scale, targeting and killing civilians based on not only political, but also ethnic and religious factors. This is commonly known as genocide. China denies using the organs of prisoners, but there's credible doubt 9 from the scientific community.

Recent evidence directly connecting executions to organ harvesting is somewhat unreliable, but I don't think China deserves the benefit of the doubt. China is a world leader in executions, and is believed to conduct more executions than the rest of the world combined. 10 Wait times for organ transplantation are extraordinarily low in China, 11 on the order of weeks — in most of the developed world these timeframes are measured in terms of years, 12 and China has been unable to explain the source for tens of thousands of transplants in the past 13 . And, looking past recent evidence, China directly admitted to using the organs of executed prisoners in 2005. 14

These atrocities are being committed against cultural minorities to further China's power. The UN published a statement in August 2018 stating that they have credible reports of over a million ethnic Uighurs being held in internment camps in Xinjiang, 15 imprisoned with various other ethnic minorities from the region. Leaks in November 2019 reported by the New York Times showed that China admits the imprisoned have committed no crimes other than dissent, 16 and that the camps were to be run with, quote, "absolutely no mercy".

It's nice to believe that we would have stood up to Nazi Germany if we had been there in the 1940's. China is our generation's chance to prove ourselves of that conviction. We talk a big game about fighting against white nationalists in our own country, and pride ourselves on standing up against "fascists". It's time we turned attention to the real fascists, on the world stage.

Instead, the staunch capitalism of America, and the West as a whole, has swooped in to leverage Chinese fascism for a profit. Marriott Hotels apologized for listing Hong Kong, Macau, and Taiwan as countries separate from China. 17 Apple removed the Taiwanese flag from iOS in China and the territories it claims. 18 Activision/Blizzard banned several players for making pro-Hong Kong statements in tournaments and online. 19 These behaviors make me ashamed to be an American.

Fuck that.

A brief history lesson: Hong Kong was originally controlled by the United Kingdom at the end of the Opium Wars. It's beyond the scope of this article, but it'll suffice to say that the United Kingdom was brutal and out of line, and the end result is that Hong Kong became a British colony. Because of this, it was protected from direct Chinese influence during China's turbulent years following, and they were insulated from the effects of the Great Leap Forward and the Cultural Revolution, which together claimed tens of millions of lives and secured the Communist Party of China's power into the present.

On July 1st, 1997, the Sino-British Joint Declaration went into effect , and Hong Kong was turned over to China. The agreement stipulated that Hong Kong would remain effectively autonomous and self-governing for a period of 50 years — until 2047. China has been gradually and

}

{

Using the right tool for the job

Drew DeVault

One of the most important choices you'll make for the software you write is what you write it in, what frameworks you use, the design methodologies to subscribe to, and so on. This choice doesn't seem to get the respect it's due. These are some of the only choices you'll make that you cannot change. Or, at least, these choices are among the most difficult ones to change.

People often question why TrueCraft is written in C# next to projects like Sway in C, alongside KnightOS in Assembly or sr.ht in Python. It would certainly be easier from the outset if I made every project in a language I'm comfortable with, using tools and libraries I'm comfortable with, and there's certainly something to be had for that. That's far from being the only concern, though.

A new project is a great means of learning a new language or framework - the only effective means, in fact. However, the inspiration and drive for new projects doesn't come often. I think that the opportunity for learning is more important than the short term results of producing working code more quickly. Making a choice that's more well suited to the problem at the expense of comfort will also help your codebase in the long run. Why squander the opportunity to choose something unfamiliar when you have the rare opportunity to start working on a new project?

I'm not advocating for you to use something new for every project, though. I'm suggesting that you detach your familiarity with your tools from the decision-making process. I often reach for old tools when starting a new project, but I have learned enough about new tools that I can judge what projects are a good use-case for them. Sometimes this doesn't work out, too - I just threw away and rewrote a prototype in C after deciding that it wasn't a good candidate for Rust.

Often it does work out, though. I'm glad I chose to learn Python for MediaCrush despite having no experience with it (thanks again for the help with that, Jose!). Today I still know it was the correct choice and knowing it has hugely expanded my programming skills, and without that choice there probably wouldn't have been a Kerbal Stuff or a sr.ht or likely even the new API we're working on at Linode. I'm glad I chose to learn C for z80e, though I had previously written emulators in C#. Without it there wouldn't be many other great tools in the KnightOS ecosystem written in C, and there wouldn't be a Sway or an aerc. I'm glad I learned ES6 and React instead of falling back on the familiar Knockout.js when building prototypes for the new Linode manager as well.

Today, I have a mental model of the benefits and drawbacks of a lot of languages, frameworks, libraries, and platforms I don't know how to use. I'm sort of waiting for projects that would be well suited to things like Rust or Django or Lisp or even Plan 9. Remember, the skills you already know

make for a great hammer, but you shouldn't nail screws to the wall.

}

{

Process scheduling and multitasking in KnightOS

Drew DeVault

I'm going to do some blogging about technical decisions made with KnightOS . It's an open-source project I've been working on for the past four years to build an open-source Unix-like kernel for TI calculators (in assembly). It's been a cool platform on top of which I can research low level systems concepts and I thought I'd share some of my findings with the world.

So, first of all, what is scheduling? For those who are completely out of the

loop, I'll explain what exactly it is and why it's necessary. Computers run on

a CPU, which executes a series of instructions in order. Each core is not

capable of running several instructions concurrently. However, you can run

hundreds of processes at once on your computer (and you probably are doing so

as you read this article). There are a number of ways of accomplishing, but the

one that suits the most situations is preemptive multitasking . This is what

KnightOS uses. You see, a CPU can only execute one instruction after another,

but you can "raise an interrupt". This will halt execution and move to some

other bit of code for a moment. This can be used to handle various events (for

example, the GameBoy raises an interrupt when a button is pressed). One of

these events is often a timer, which raises an interrupt at a fixed interval.

This is the mechanism by which preemptive multitasking is accomplished.

Let's say for a moment that you have two programs loaded into memory and running, at addresses 0x1000 and 0x2000. Your kernel has an interrupt handler at 0x100. So if program A is running and an interrupt fires, the following happens:

0x1000 is pushed to the stack as the return address

The program counter is set to 0x100 and the interrupt runs

The interrupt concludes and returns, which pops 0x1000 from the stack and into the program counter.

Once the interrupt handler runs, however, the kernel has a chance to be sneaky:

0x1000 is pushed to the stack as the return address

The program counter is set to 0x100 and the interrupt runs

The interrupt removes 0x1000 from the stack and puts 0x2000 there instead

The interrupt concludes and returns, which pops 0x2000 from the stack and into the program counter.

Now the interrupt has switched the CPU from program A to program B. And the next time an interrupt occurs, the kernel can switch from program B to program A. This event is called a "context switch". This is the basis of preemptive multitasking. On top of this, however, there are lots of ideas around which processes should get CPU time and when. Some systems have more complex schedulers, but KnightOS runs on limited hardware and I wanted the context switch to be short and sweet so that the running processes get as much of the CPU as possible. I'll explain the simple KnightOS scheduling algorithm here. First, its goals:

Short and simple context switches

Ability to suspend processes when not in foreground

Ability to run background processes

What KnightOS uses is a simple round robin with the ability to suspend threads. That is, we have a list of processes and then some flags, among which is whether or not the processes is currently suspended. So say we have this list of processes in memory:

1: PC=0x2000, not suspended

2: PC=0x2000, not suspended

3: PC=0x2000, suspended

4: PC=0x2000, not suspended

As process 1 is running and an interrupt fires, the kernel looks at this table and picks the next non-suspended process to run - process 2. On the next interrupt, it does it again, skipping process 3 and giving time to process 4.

To actually implement this, we have to think about the stack. KnightOS runs on z80 processors, which have a single stack and a shared memory space. The CPU uses the PC register to keep track of which address the current instruction is at. That is, say you compile this code:

ld a, 10
inc a
ld (hl), a

This compiles to the machine code 3E 0A 3C 77. Say we load this program at 0x8000 - then 0x8000 will point to ld a, 10 . When the CPU finishes executing this instruction, it advances PC to 0x8002 (since ld a, 10 is a two-byte instruc-

```
}
```

```
standard-article( title: [Updating to
the Latest Ansible on Ubuntu], au-
thor: [Scott Johnson (fuzzyblog)],
source-name: [Scott Johnson (fuzzy-
blog)], images: (), paragraphs: ( [One
of the tricky bits about Ansible
is that new features in the “lan-
guage” always require the latest ver-
sion of Ansible itself. And, while
logical, this can easily bite you.
Here’s an example:], [class=“high-
light”> TASK [deploy_hate-language-
modeling_systemd_start : just force
systemd to re-execute itself (2.8 and
above)] *** fatal : [ aws_master2 ]:
FAILED! => {“changed” : false ,
“msg” : “Unsupported parameters for
(systemd) module: daemon_reexec
Supported parameters include:
daemon_reload, enabled, masked,
name, no_block, state, user” } to retry,
use : -limit @/home/ubuntu/ansi-
ble/playbook_deploy_hate-language-
modeling_master.retry], [What that
translates to is that Ansible itself
doesn’t know how to use the parame-
ter daemon_reexec . And when you’ve
been using an Ansible feature success-
fully and then it doesn’t work on the
server, that usually is a rock solid
signal “update ansible”.], [Here’s how
to do this on Ubuntu:], ), insert-map:
(:), word-count: 155, edited-for-length:
false, debug-mode: false, )
```

```
standard-article( title: [A Week of
Whole 30 Meals From My Recipes],
author: [Scott Johnson (fuzzyblog)],
source-name: [Scott Johnson (fuzzy-
blog)], images: (), paragraphs: ( [My el-
dest son needs to lose about 15 pounds
before he leaves for boot camp in the
Marines. So here is a week of Whole 30
Meals planned out for him. Here are
more details on Whole 30 (scroll down
about 1/2 way)]., [id=“monday”>Mon-
day], [Breakfast: 3 Squishy Eggs , Black
Coffee], [Lunch: Hard boiled eggs,
nuts, fruit], [Dinner: Grilled Chicken
Breast, Baked potato with Olive Oil (no
dairy), Salad with homemade dress-
ing], [Snack: handful of cashews],
[Sweet: Lara bar], [Reward: Dates],
[Beverage: LaCroix Water], [id=“tues-
day”>Tuesday], [Breakfast: 2 Fried
Eggs], [Lunch: Hard boiled eggs, nuts,
fruit], [Dinner: Garlicky Pork Chops,
Roasted Carrots , Salad], [Snack:
Mixed Nuts; no Peanuts], [Sweet:
Lara bar], [Reward: Dates], [Bev-
erage: LaCroix Water], [id=“wednes-
day”>Wednesday], [Breakfast: Scram-
bled Eggs], [Lunch: Burger patty,
nuts, fruit], [Dinner: Steak, Broc-
coli, Salad], [Snack: Terra Chips],
[Sweet: Lara bar], [Reward: Apple],
[Beverage: LaCroix Water], [id=“thurs-
day”>Thursday], [Breakfast: Omelet],
[Lunch: Hard boiled eggs, nuts,
fruit], [Dinner: Burger, Hash Browns],
[Snack: Cashews, Mixed Nuts], [Sweet:
Lara Bar], [Reward: Berries], [Bev-
erage: LaCroix Water], [id=“friday”>Fri-
day], [Breakfast: Scrambled Eggs],
[Lunch: Hard boiled eggs, nuts,
fruit], [Dinner: Steak, Left Over Hash
Browns, Sautéed Bell Peppers and
Onions], [Snack: Lara Bar], [Sweet:
Berries], [Reward: Dates], [Beverage:
LaCroix Water], [id=“saturday”>Satur-
day], [Breakfast: Omelet], [Lunch:
Burgers Wrapped in Lettuce], [Din-
ner: Left Over Steak, Mashed Potatoes
Made with Chicken Broth], [Snack:
Cashews], [Sweet: Lara Bar], [Reward:
Berries], [Beverage: LaCroix Wa-
ter], [id=“sunday”>Sunday], [Break-
fast: Fried Eggs], [Lunch: Hard Boiled
Eggs, Salad], [Dinner: Meat Loaf,
Mashed Potatoes], [Snack: Lara Bar],
[Sweet: Lara Bar], [Reward: Berries],
[Beverage: LaCroix Water], ), insert-
map: (:), word-count: 250, edited-for-
length: false, debug-mode: false, )
```

```
brief-group(( brief-item([Axel
Rauschmayer (2ality)], source-
name: [Axel Rauschmayer (2ality)],
[TypeScript’s satisfies operator lets us
check the type of a value (mostly) with-
out influencing it. In this blog post,
we examine how exactly it works and
where it’s useful.]
```

```
brief-item([Thibaud Esnouf], source-
name: [Drivy / Getaround Tech], [At
Drivy, we have defined our very own
design system. This system describes
our visual guidelines and rules, and is
composed of visual web components.
For each component, we have created
a React implementation that could
easily be used by our design team
to build a web site documentation
(thanks to MDX ).])
```

```
brief-item([Axel Rauschmayer
(2ality)], source-name: [Axel
Rauschmayer (2ality)], [In this blog
post, we look at how WebAssembly
has become an ecosystem for many
programming languages and what
technologies enable that.]
```

```
brief-item([The Hacker News], source-
name: [The Hacker News (cyberse-
curity)], [Two more GitHub Actions
workflows have become the latest to
be compromised by credential-steal-
ing malware by a threat actor known
as TeamPCP, the cloud-native cyber-
criminal operation also behind the
Trivy supply chain attack. The work-
flows, both maintained by the supply
chain security company Checkmarx,
are listed below -
```

```
checkmarx/ast-github-action check-
marx/kics-github-action
```

```
Cloud security))
```

```
brief-item([Scott Johnson (fuzzyblog)],
source-name: [Scott Johnson (fuzzy-
blog)], [class=“center”>
```

```
Your Rails 6 master key is located in:
```

```
class=“language-plaintext highlighter-
rouge”>
```

```
class=“highlight”> config/credentials/
production.key
```

```
and it is likely stored as a hidden file. I
believe that this is generated when the
project is generated but I’m not 100%
certain.
```


See this Reddit post .])

brief-item([Axel Rauschmayer (2ality)], source-name: [Axel Rauschmayer (2ality)], [In this blog post, we explore how we can style text that we log to the console in Node.js.

Some of the examples use a Unix shell but most of the code should also work on Windows.])

brief-item([The Hacker News], source-name: [The Hacker News (cybersecurity)], [Some weeks in security feel loud. This one feels sneaky. Less big dramatic fireworks, more of that slow creeping sense that too many people are getting way too comfortable abusing things they probably shouldn't even be touching. There's a little bit of everything in this one, too. Weird delivery tricks, old problems coming back in slightly worse forms, shady infrastructure doing])

brief-item([Axel Rauschmayer (2ality)], source-name: [Axel Rauschmayer (2ality)], [Starting with v23.6.0 , Node.js supports TypeScript without any flags. This blog post explains how it works and what to look out for.])

brief-item([Faouz EL FASSI], source-name: [Drivy / Getaround Tech], [In the first of this series of blog posts about Data-Warehousing, I've been talking about how we use and manage our Amazon Redshift cluster at Drivy.])

))

The New Mirror

Vol. 1, No. 011 · 2026-03-30

Curated and composed automatically.